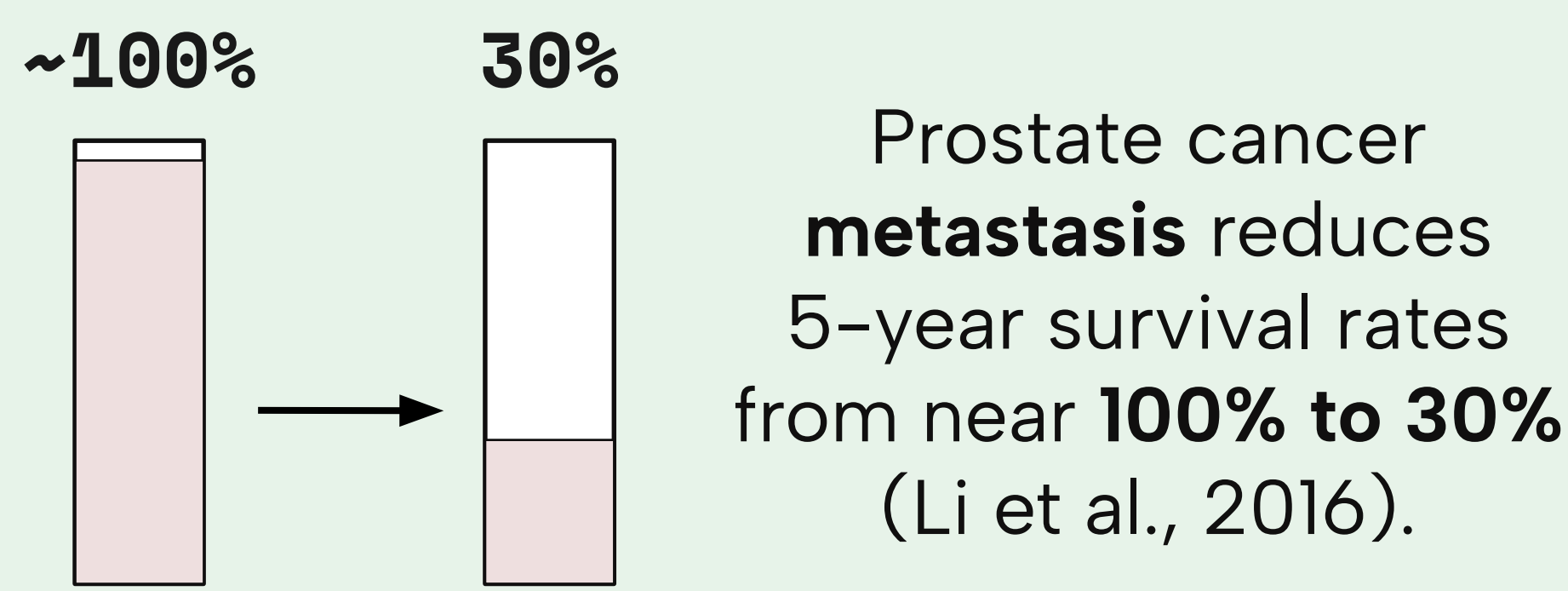


INHIBITION OF THE EPITHELIAL-MESENCHYMAL TRANSITION OF HUMAN PROSTATE CANCER CELLS USING IBUPROFEN

INTRODUCTION



Metastasis = migration of tumor cells, facilitated by **EMT**, spreading cancer

Epithelial-Mesenchymal Transition (EMT)

EPITHELIAL CELLS → MESENCHYMAL CELLS

- Forms outer layer of bodily surfaces.
- Adhesive/polarized.
- Migratory** and invasive.
- In cancer, creates new tumors.

Initial Causes: developmental signaling → **inflammation** + damage (Kalluri & Weinberg, 2009)

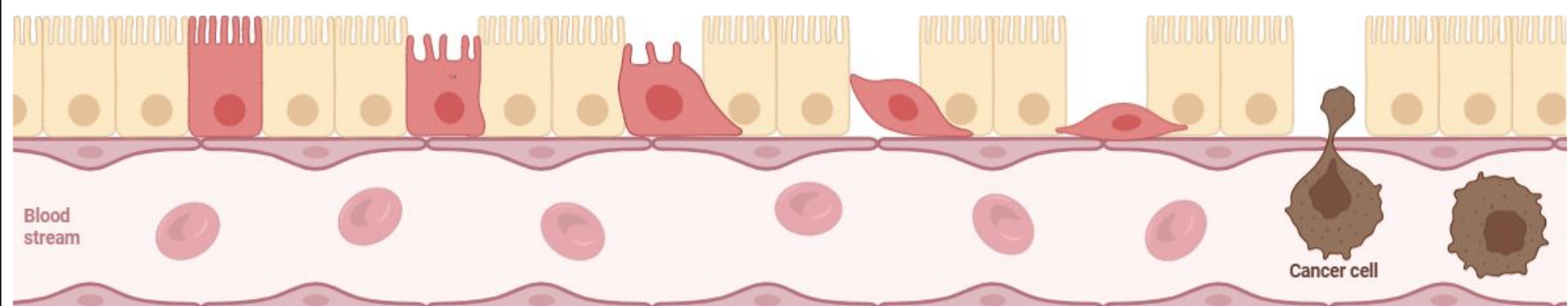


Fig 1: Cells undergoing the epithelial-mesenchymal transition become migratory and lead to metastasis. Figure created in BioRender by researcher.

Ibuprofen = NSAID =

Blocks Cyclooxygenase (Song et al., 2019)

NonSteroidal Anti-Inflammatory
Drugs (NSAIDs)

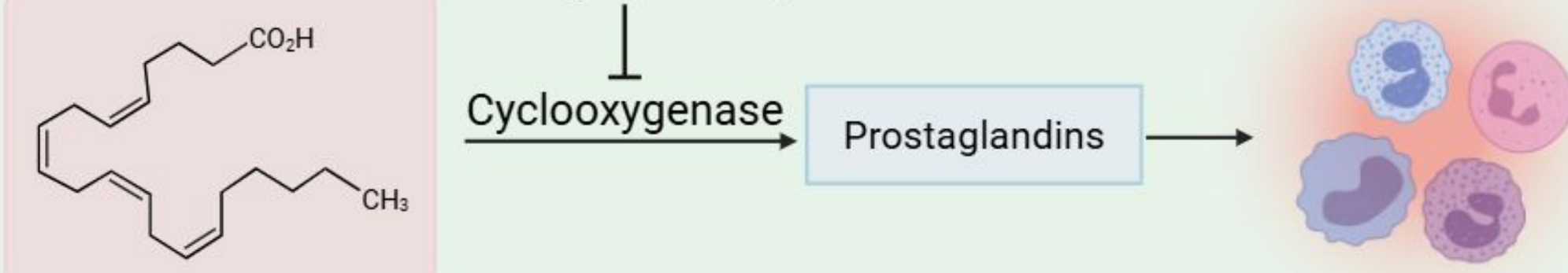


Fig 2: Ibuprofen blocks cyclooxygenase in inflammation. Figure created in BioRender by researcher.

HYPOTHESIS

If the amount of **ibuprofen** in the cells increases, then the amount of **epithelial-mesenchymal transition and migration** will decrease.

Variables

IV: Ibuprofen (0, 300, 600, 1200 μM)

DV: Cell migration

Control: Negative control (99% F12K, 1% FBS)

PURPOSE

To help treatment of prostate cancer by **reducing tumor metastasis**.

METHODS

Normoxia(no EMT)	Hypoxia(induced EMT)
Negative Control (0 μM)	Negative Control (0 μM)
300 μM Ibuprofen	300 μM Ibuprofen
600 μM Ibuprofen	600 μM Ibuprofen
1200 μM Ibuprofen	1200 μM Ibuprofen

SCRATCH ASSAY - to measure migration

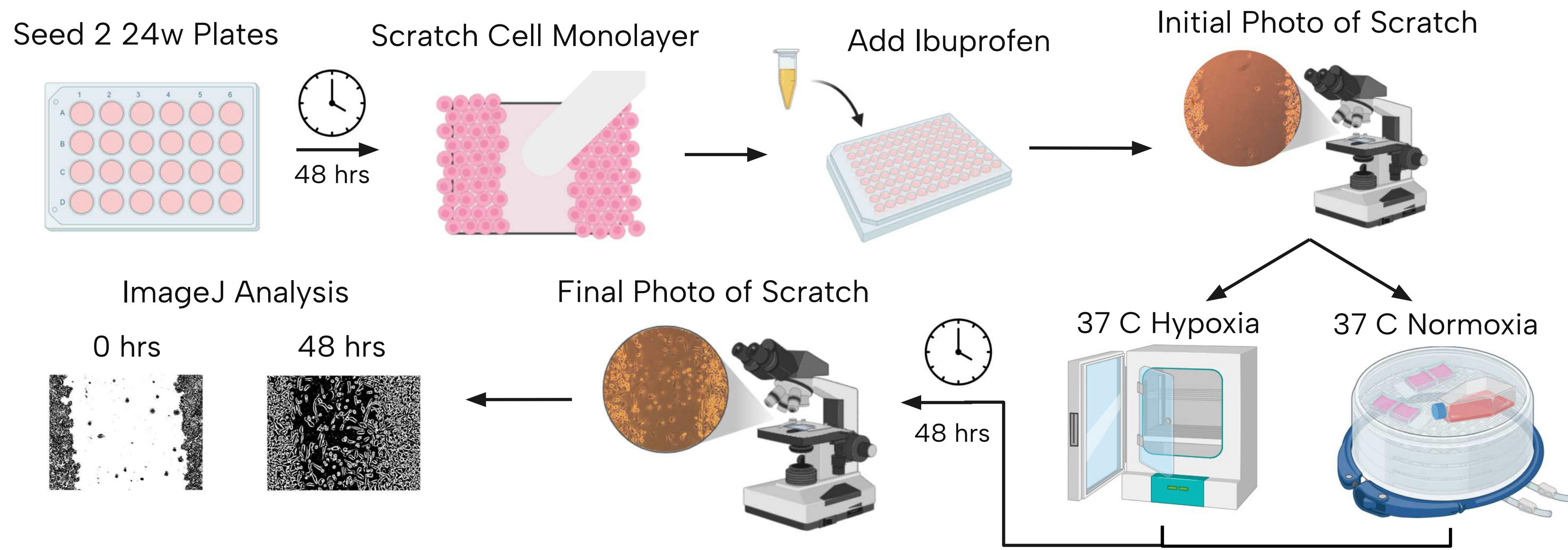


Fig 3: Scratch Assay Method. Figure created in BioRender by researcher.

- Two 24-well plates are seeded and incubated at 37 C for 48 hours, until at >85% confluency.
- Growth media is removed from both plates and are washed with PBS.
- Fresh PBS is added and each well is scratched with a pipette tip.
- PBS is removed and ibuprofen treatments dissolved in F12K, 1% FBS are added.
- Photos of wells are taken at 0 hours and plates are incubated in either 37 C normoxia or 37 C hypoxia for 48 hours, when 48 hour photos are taken.
- The areas of the images are quantified using ImageJ. (n = 8 replicates per group)

RESULTS

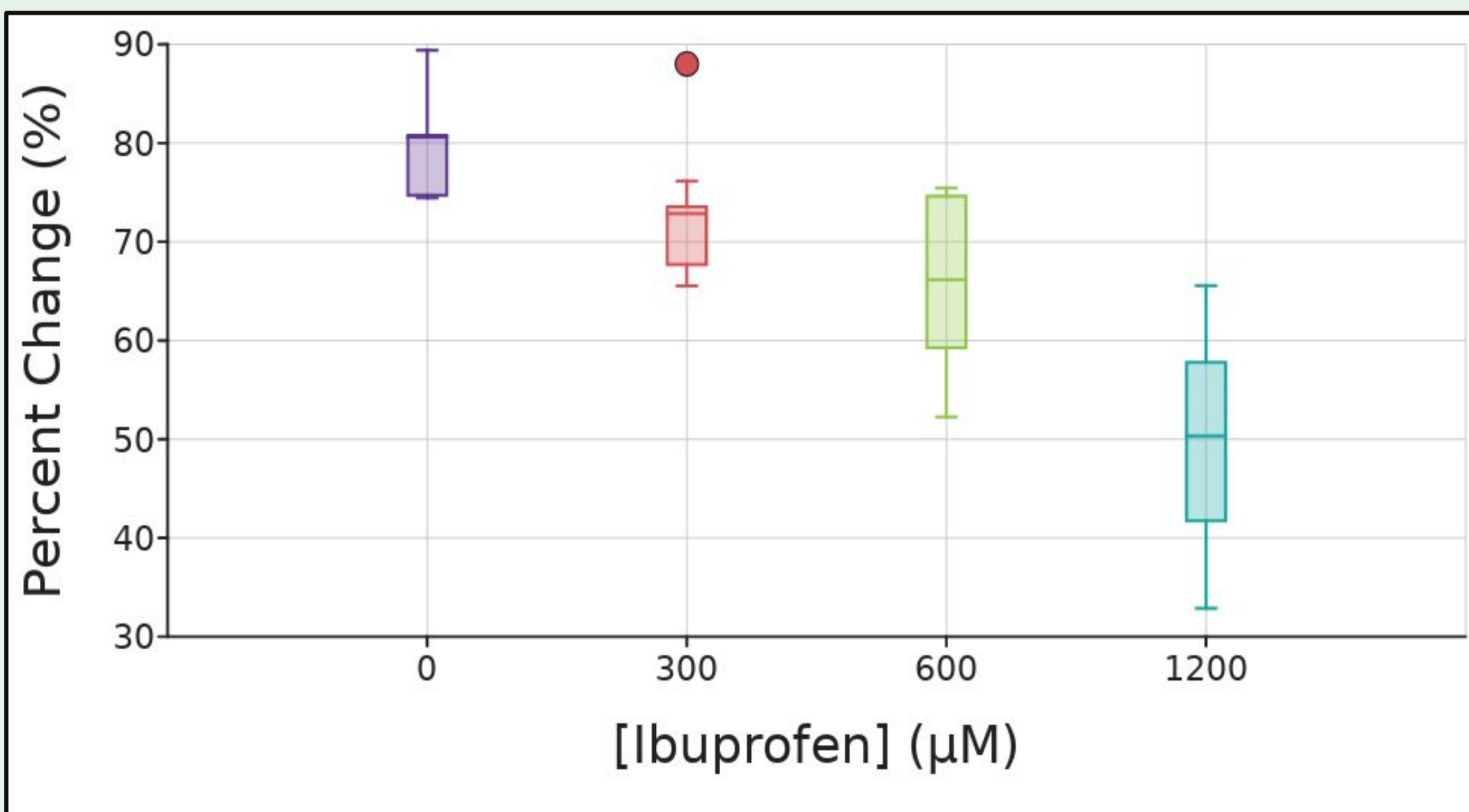


Fig 4: Ibuprofen Concentration vs. Percent Change in Area of Scratch After **Hypoxia**. Figure created in DataClassroom by researcher.

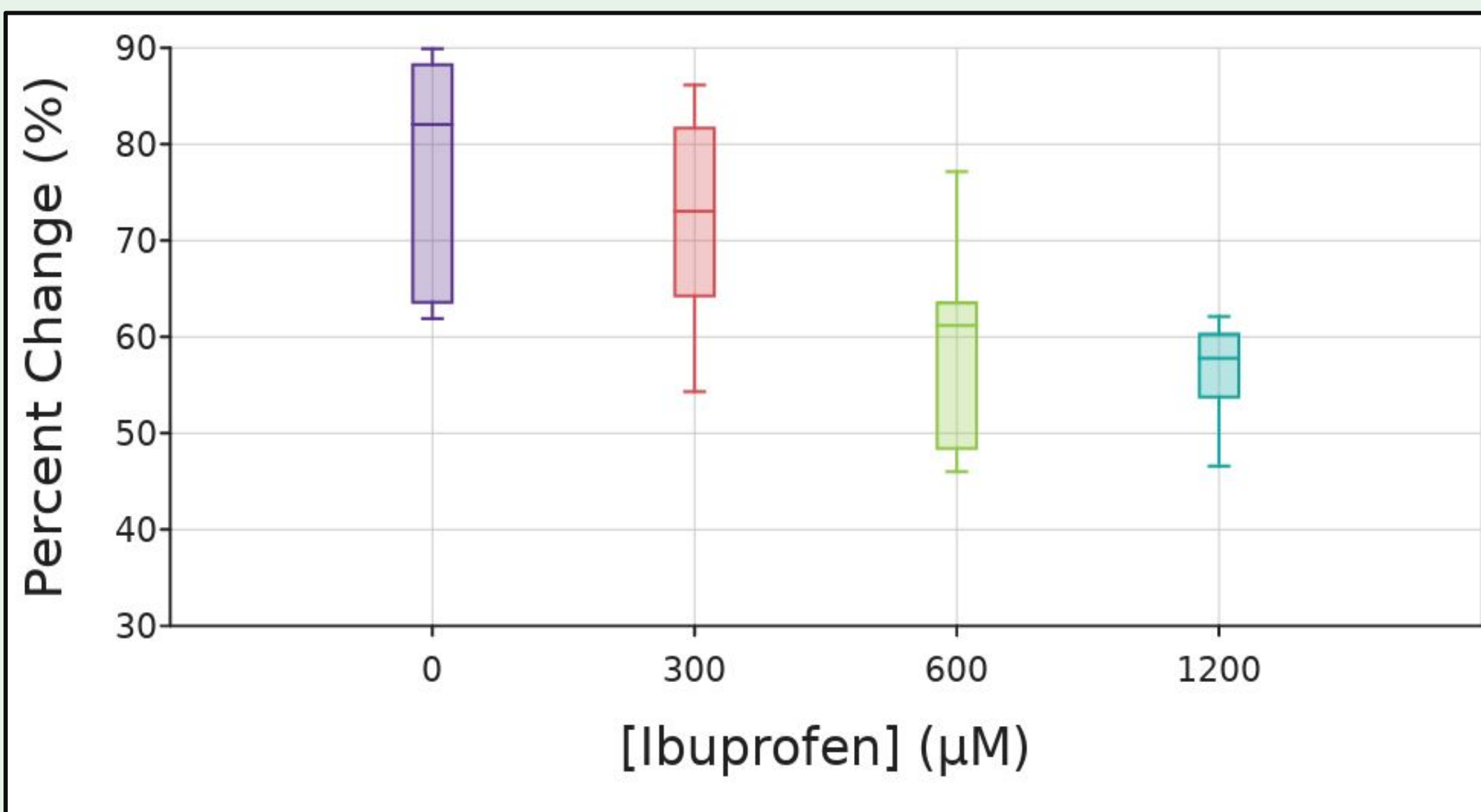
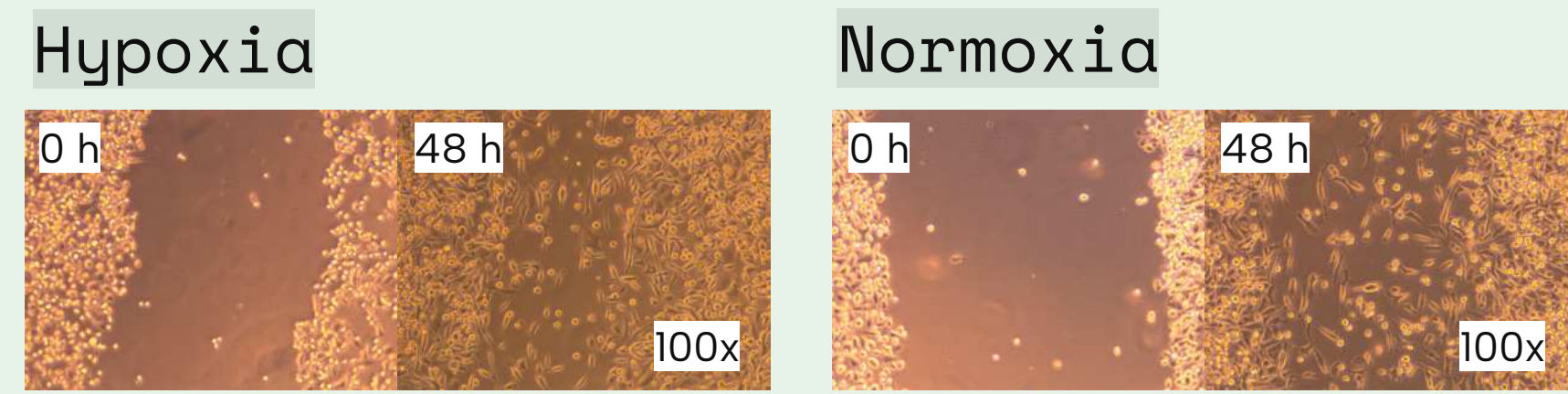


Fig 5: Ibuprofen Concentration vs. Percent Change in Area of Scratch After **Normoxia**. Figure created in DataClassroom by researcher.

CONCLUSION



Pair	P-Value	KW P-Value
0 μM - 300 μM	0.66	<0.01 (Significantly Different)
0 μM - 600 μM	0.07	
0 μM - 1200 μM	<0.01	
300 μM - 600 μM	1.00	
300 μM - 1200 μM	0.01	
600 μM - 1200 μM	0.29	

Table 1: Table created in Google Slides by Researcher.

Pair	P-Value	KW P-Value
0 μM - 300 μM	1.00	<0.01 (Significantly Different)
0 μM - 600 μM	0.04	
0 μM - 1200 μM	<0.01	
300 μM - 600 μM	0.29	
300 μM - 1200 μM	0.03	
600 μM - 1200 μM	1.00	

Table 2: Table created in Google Slides by Researcher.

Significant decrease of percent change in area with Ibuprofen. → Ibuprofen can reduce migration in cells.

Pair	P-Value	KW P-Value
0 Hpx μM - 0 Nmx μM	1.00	1.00 (Not Significantly Different)
300 μM Hpx - 300 μM Nmx	1.00	
600 μM Hpx - 600 μM Nmx	1.00	
1200 μM Hpx - 1200 μM Nmx	1.00	

No significant difference in migration between Hypoxia and Normoxia.

Not enough evidence to conclude EMT is occurring.

Table 3: Table created in Google Slides by Researcher.

- Data Supports Hypothesis
- An **increase in Ibuprofen** in the cells causes a significant decrease in percent change in area of the scratch → indicating **decreased migration**
- Unlike what was expected, there was not enough evidence to conclude that there is a difference in migration between hypoxia and normoxia. → **Do not have evidence to support EMT occurring** in the cells.
- There is evidence to support that ibuprofen treatments may be able to reduce the instance of prostate cancer metastasis, helping its treatment; however, none supporting the reduction of EMT.

FUTURE RESEARCH

Future work could involve directly measuring Epithelial-Mesenchymal transition expression in the cells, in relation to concentration of ibuprofen, to solidify their relationship.

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